

Unramifiedness of $\mathcal{P}^{[d]G}$ at the exceptional divisor

(standalone excerpt)

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Roadmap. [Section 1](#) collects, without re-deriving them, the facts about symmetric powers of torsors that the proof uses, stated over an abstract base. [Section 2](#) specialises this to a smooth projective curve: it recalls the notion of bounded ramification and fixes the geometric data (the symmetric tower over $U^{(d)}$, the compactification $C' \rightarrow C$, the blowups at the points at infinity). [Section 3](#) assembles everything into a single master diagram, and [Section 4](#) states and proves the unramifiedness theorem.

1 Background

Throughout this section S is a base scheme, G a finite abelian group, $H \subseteq G$ a subgroup, and $d \geq 0$ an integer. All bases are assumed Zariski-locally quasi-projective over S , so that the symmetric-power quotients below exist as schemes.

1.1 Symmetric powers of torsors

Let $\mathcal{P}$ be a G -torsor over an S -scheme X . For each $i \in \llbracket 1, d \rrbracket$ write $p_i : X^{\times sd} \rightarrow X$ for the i -th projection and consider the G -torsor

$$p_1^{-1}\mathcal{P} \otimes^G \dots \otimes^G p_d^{-1}\mathcal{P} = K_G \backslash \mathcal{P}^{\times sd}$$

over $X^{\times sd}$, where $K_G \subseteq G^d$ is the kernel of the multiplication morphism $G^d \rightarrow G$ (the dependence on d is suppressed in the notation). The symmetric group S_d acts on the right on $K_G \backslash \mathcal{P}^{\times sd}$ by $(p_i)_i \cdot \sigma = (p_{\sigma(i)})_i$, and this action commutes with the left G -action.

Proposition 1 ([\[Gui19\]](#), Proposition 2.32). *The right action of S_d on $K_G \backslash \mathcal{P}^{\times sd}$ is admissible, so that the quotient $\mathcal{P}^{[d]}$ of $K_G \backslash \mathcal{P}^{\times sd}$ by S_d exists as a scheme over S . Moreover, the canonical morphism $\mathcal{P}^{[d]} \rightarrow \mathrm{Sym}_S^d(X)$ is a G -torsor, and the morphism*

$$p_1^{-1}\mathcal{P} \otimes^G \dots \otimes^G p_d^{-1}\mathcal{P} \rightarrow r^{-1}\mathcal{P}^{[d]},$$

where $r : X^{\times sd} \rightarrow \mathrm{Sym}_S^d(X)$ is the canonical projection, is an isomorphism of G -torsors over $X^{\times sd}$.

When G is abelian the condition $\prod_{i=1}^d g_i = 1$ defining K_G is invariant under permuting the coordinates, so the (left) S_d -action normalises the K_G -action. Writing $\Xi = K_G \rtimes S_d$ for the resulting finite group, acting on $\mathcal{P}^{\times s^d}$ by

$$((g_1, \dots, g_d), \sigma) \star (p_1, \dots, p_d) = (g_1 p_{\sigma^{-1}(1)}, \dots, g_d p_{\sigma^{-1}(d)}),$$

one has $(K_G \backslash \mathcal{P}^{\times s^d})/S_d \cong \Xi \backslash \mathcal{P}^{\times s^d}$.

Corollary 2. *There is a natural canonical morphism*

$$q : S_d \backslash \mathcal{P}^{\times s^d} \longrightarrow \Xi \backslash \mathcal{P}^{\times s^d} \quad \text{equivalently} \quad q : S_d \backslash \mathcal{P}^{\times s^d} \longrightarrow (K_G \backslash \mathcal{P}^{\times s^d})/S_d,$$

the comparison between the symmetric power of $\mathcal{P}$ as a scheme and the symmetric power of $\mathcal{P}$ as a G -torsor. We write $\mathcal{P}^{(d)}$ for the former and $\mathcal{P}^{[d]}$ for the latter, so that $q : \mathcal{P}^{(d)} \rightarrow \mathcal{P}^{[d]}$.

Proof. S_d is a subgroup of Ξ , hence any Ξ -invariant morphism is automatically S_d -invariant. $\square$

1.2 The canonical decomposition of a torsor

Proposition 3. *Let $\mathcal{P} \rightarrow X$ be a G -torsor in $X_{\text{ét}}$ and let $H \subseteq G$ be a subgroup, with quotient map $\pi : G \rightarrow G/H$. Then $\mathcal{P} \rightarrow X$ factors*

$$\mathcal{P} \xrightarrow{\phi} \mathcal{P}_{G/H} \xrightarrow{\psi} X,$$

uniquely up to a unique isomorphism commuting with the projections, where

1. $\mathcal{P}_{G/H} := \mathcal{P} \times^G (G/H)$ is a G/H -torsor over X , and
2. $\mathcal{P}$ is an H -torsor over the scheme $\mathcal{P}_{G/H}$ via ϕ .

1.3 The tower of symmetric quotients

Fix a G -torsor $\mathcal{P} \rightarrow X$ and a subgroup $H \subseteq G$, with canonical factorisation $\mathcal{P} \xrightarrow{\phi} \mathcal{P}_{G/H} \xrightarrow{\psi} X$ as in [Proposition 3](#) (ϕ an H -torsor, ψ a G/H -torsor). We record how the symmetric-power construction of [Section 1.1](#) interacts with this factorisation.

Summation kernels. Besides $K_G = \ker(G^d \rightarrow G)$ we need

$$K_H := \ker(H^d \rightarrow H), \quad K_{G/H} := \ker((G/H)^d \rightarrow G/H).$$

The snake lemma applied to

$$\begin{array}{ccccccc} 0 & \longrightarrow & H^d & \longrightarrow & G^d & \longrightarrow & (G/H)^d \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & H & \longrightarrow & G & \longrightarrow & G/H \longrightarrow 0, \end{array}$$

(the vertical maps being the surjective summation morphisms) gives a short exact sequence

$$0 \rightarrow K_H \rightarrow K_G \rightarrow K_{G/H} \rightarrow 0, \tag{1}$$

hence $K_G/K_H \cong K_{G/H}$. Inside G^d we have the chains of subgroups

$$K_H \subseteq K_G \subseteq H^d + K_G, \quad H^d \subseteq H^d + K_G.$$

Four quotient schemes. Let S_d act on $\mathcal{P}^{\times s^d}$ as in [Corollary 2](#), and let each of the subgroups above act diagonally via the G -action on $\mathcal{P}$. Set

$$\begin{aligned}\mathcal{P}^{[d]_G} &:= (K_G \rtimes S_d) \backslash \mathcal{P}^{\times s^d}, & \text{the symmetric } G\text{-power of } \mathcal{P}; \\ \mathcal{P}^{[d]_H} &:= (K_H \rtimes S_d) \backslash \mathcal{P}^{\times s^d}, & \text{the symmetric } H\text{-power of } \mathcal{P} \text{ relative to } \phi; \\ (\mathcal{P}_{G/H})^{(d)} &:= (H^d \rtimes S_d) \backslash \mathcal{P}^{\times s^d}, & \text{the symmetric power of the scheme } \mathcal{P}_{G/H}; \\ (\mathcal{P}_{G/H})^{[d]} &:= ((H^d + K_G) \rtimes S_d) \backslash \mathcal{P}^{\times s^d}, & \text{the symmetric } G/H\text{-power of } \mathcal{P}_{G/H}.\end{aligned}$$

Proposition 4. *The four quotient schemes above exist and fit into a canonical commutative diagram*

$$\begin{array}{ccc}\mathcal{P}^{[d]_H} & \xrightarrow{H} & (\mathcal{P}_{G/H})^{(d)} \\ \downarrow & & \downarrow q \\ \mathcal{P}^{[d]_G} & \xrightarrow{H} & (\mathcal{P}_{G/H})^{[d]} \\ & \searrow G & \downarrow G/H \\ & & X^{(d)},\end{array}$$

in which each labelled arrow is a torsor under the indicated constant group, the right-hand vertical q is the canonical projection of [Corollary 2](#) applied to $\mathcal{P}_{G/H} \rightarrow X$, and the upper square is Cartesian:

$$\mathcal{P}^{[d]_H} \cong \mathcal{P}^{[d]_G} \times_{(\mathcal{P}_{G/H})^{[d]}} (\mathcal{P}_{G/H})^{(d)}.$$

The unlabelled left-hand vertical $\mathcal{P}^{[d]_H} \rightarrow \mathcal{P}^{[d]_G}$ is then the base change of q along $\mathcal{P}^{[d]_G} \rightarrow (\mathcal{P}_{G/H})^{[d]}$. Neither q nor this left-hand vertical is, in general, a torsor under a constant group scheme.

Proof. The four labelled arrows are torsors under the indicated constant groups by applying [Proposition 1](#) and [Proposition 3](#):

- to the G -torsor $\mathcal{P} \rightarrow X$, giving the G -torsor $\mathcal{P}^{[d]_G} \rightarrow X^{(d)}$ (the diagonal arrow);
- to the H -torsor $\phi : \mathcal{P} \rightarrow \mathcal{P}_{G/H}$ (with $\mathcal{P}_{G/H}$ as base), giving the H -torsor $\mathcal{P}^{[d]_H} \rightarrow (\mathcal{P}_{G/H})^{(d)}$ (the top arrow);
- to the G/H -torsor $\psi : \mathcal{P}_{G/H} \rightarrow X$, giving the G/H -torsor $(\mathcal{P}_{G/H})^{[d]} \rightarrow X^{(d)}$ (the bottom-right vertical) and, via [Corollary 2](#), the comparison map $q : (\mathcal{P}_{G/H})^{(d)} \rightarrow (\mathcal{P}_{G/H})^{[d]}$.

The factorisation $\mathcal{P}^{[d]_G} \xrightarrow{H} (\mathcal{P}_{G/H})^{[d]} \xrightarrow{G/H} X^{(d)}$ of the diagonal arrow is the canonical decomposition of [Proposition 3](#) applied to $\mathcal{P}^{[d]_G} \rightarrow X^{(d)}$ and $H \subseteq G$: both $\mathcal{P}^{[d]_G} \times^G (G/H)$ and $(\mathcal{P}_{G/H})^{[d]}$ are realised as the admissible quotient $((H^d + K_G) \rtimes S_d) \backslash \mathcal{P}^{\times s^d}$.

For the Cartesian property, set $F := \mathcal{P}^{[d]_G} \times_{(\mathcal{P}_{G/H})^{[d]}} (\mathcal{P}_{G/H})^{(d)}$. Base-changing the H -torsor $\mathcal{P}^{[d]_G} \rightarrow (\mathcal{P}_{G/H})^{[d]}$ along q exhibits F as an H -torsor over $(\mathcal{P}_{G/H})^{(d)}$. The universal property of the fibre product gives a canonical H -equivariant morphism $\mathcal{P}^{[d]_H} \rightarrow F$ over $(\mathcal{P}_{G/H})^{(d)}$; any morphism of H -torsors is an isomorphism, so $\mathcal{P}^{[d]_H} \cong F$ and the upper square is Cartesian. Via this square the left vertical $\mathcal{P}^{[d]_H} \rightarrow \mathcal{P}^{[d]_G}$ is identified with the base change of q along $\mathcal{P}^{[d]_G} \rightarrow (\mathcal{P}_{G/H})^{[d]}$. $\square$

2 The geometric setting

2.1 Bounded ramification along a curve

Let C be a smooth projective geometrically connected curve over a field k , let $\mathfrak{m}$ be a fixed modulus, and put $U = C \setminus \mathfrak{m}$. Let $\mathcal{P} \rightarrow U$ be a G -torsor with G finite abelian. For a closed point P of C , passing to the completion $\widehat{\mathcal{O}}_{C,P}$ gives the complete local field $\widehat{K}_P$, and restricting $\mathcal{P}$ to $\mathrm{Spec}(\widehat{K}_P)$ yields a G -torsor over a field, hence a disjoint union of spectra of finite separable field extensions

$$\mathcal{P}|_{\mathrm{Spec}(\widehat{K}_P)} \cong \bigsqcup \mathrm{Spec}(F),$$

in which the extensions $F/\widehat{K}_P$ are all isomorphic and Galois, $H = \mathrm{Gal}(F/\widehat{K}_P)$ is the stabiliser of a connected component (a subgroup of G), and there are $[G : H]$ components. Write H^r for the r -th higher ramification group of $F/\widehat{K}_P$.

Definition 5. The extension $F/\widehat{K}_P$ has *ramification bounded by r* if the ramification group H^r is trivial. We say $\mathcal{P}$ has *ramification bounded by r at P* if the corresponding local extensions $F/\widehat{K}_P$ satisfy this condition.

2.2 The symmetric tower over $U^{(d)}$

For the rest of the note we fix, in addition to the data above:

- a normal subgroup $H \trianglelefteq G$;
- a point $p_\infty \in \mathfrak{m}$ and an integer $d \geq 1$;

and we assume that $\mathcal{P} \rightarrow U$ has ramification bounded by d at p_∞ in the sense of [Definition 5](#) (i.e. the d -th higher ramification group of the extension of DVRs at p_∞ is trivial).

Apply the construction of [Section 1.3](#) with $S = \mathrm{Spec} k$ and base $X = U$. This produces the four quotient schemes

$$\mathcal{P}^{[d]_H}, \quad \mathcal{P}^{[d]_G}, \quad (\mathcal{P}_{G/H})^{(d)}, \quad (\mathcal{P}_{G/H})^{[d]},$$

and, by [Proposition 4](#), the canonical commutative diagram with Cartesian upper square

$$\begin{array}{ccc} \mathcal{P}^{[d]_H} & \xrightarrow{H} & (\mathcal{P}_{G/H})^{(d)} \\ \downarrow & & \downarrow q \\ \mathcal{P}^{[d]_G} & \xrightarrow{H} & (\mathcal{P}_{G/H})^{[d]} \\ & \searrow G & \downarrow G/H \\ & & U^{(d)}, \end{array}$$

in which the arrows labelled H , G/H and G are torsors under the indicated finite constant groups (in particular, étale).

2.3 Compactification and blowup

The G/H -torsor $\mathcal{P}_{G/H} \rightarrow U$ compactifies to a finite morphism of smooth projective curves $f: C' \rightarrow C$; fix $p'_\infty \in f^{-1}(p_\infty)$. Taking d -fold symmetric powers gives the finite morphism

$$f^{(d)}: C'^{(d)} \rightarrow C^{(d)}, \quad d \cdot p'_\infty \mapsto d \cdot p_\infty.$$

Let

$$\pi_C: X_C \rightarrow C^{(d)}, \quad \pi_{C'}: X_{C'} \rightarrow C'^{(d)}$$

be the blowups at $d \cdot p_\infty$ and $d \cdot p'_\infty$ respectively, with exceptional divisors $E_C, E_{C'}$ and generic points $\eta_C, \eta_{C'}$. The local rings at these generic points are the discrete valuation rings

$$V_C := \mathcal{O}_{X_C, \eta_C} \subset K(C^{(d)}), \quad V_{C'} := \mathcal{O}_{X_{C'}, \eta_{C'}} \subset K(C'^{(d)}).$$

3 The master diagram

The torsor tower over $U^{(d)}$ embeds into the projective geometry of $C^{(d)}$ and $C'^{(d)}$, which in turn carry the canonical blowups at the points at infinity. Assembling all of this:

$$\begin{array}{ccccccc}
 \mathcal{P}^{[d]_H} & \xrightarrow{H} & (\mathcal{P}_{G/H})^{(d)} & \xhookrightarrow{\iota'} & C'^{(d)} & \xleftarrow{\pi_{C'}} & X_{C'} \\
 \downarrow & & \downarrow q & & \downarrow f^{(d)} & & \downarrow f_X \\
 \mathcal{P}^{[d]_G} & \xrightarrow{H} & (\mathcal{P}_{G/H})^{[d]} & & & & \\
 & \searrow G & \downarrow G/H & & & & \\
 & & U^{(d)} & \xhookrightarrow{\iota} & C^{(d)} & \xleftarrow{\pi_C} & X_C
 \end{array}$$

Reading the diagram from left to right:

Left block (columns 1–2): the torsor tower of [Proposition 4](#) over $U^{(d)}$ — H -torsors horizontally, the G/H -torsor on the lower right, and the composite G -torsor $\mathcal{P}^{[d]_G} \rightarrow U^{(d)}$.

Middle block (column 3): the compactifications. Here ι is the canonical open immersion $U^{(d)} \hookrightarrow C^{(d)}$, and ι' is the open immersion $(\mathcal{P}_{G/H})^{(d)} \hookrightarrow C'^{(d)}$ coming from $\mathcal{P}_{G/H} \subset C'$. Restricting $f^{(d)}$ along ι' recovers the canonical map $(\mathcal{P}_{G/H})^{(d)} \rightarrow U^{(d)}$, i.e. the composite of q with the G/H -torsor.

Right block (column 4): the blowups at the diagonal points at infinity. The dashed arrow $f_X: X_{C'} \dashrightarrow X_C$ is the rational lift of $f^{(d)}$, defined on the strict transform; on generic points of the exceptional divisors it sends $\eta_{C'} \mapsto \eta_C$, inducing an extension of DVRs $V_C \hookrightarrow V_{C'}$.

4 Unramifiedness of $\mathcal{P}^{[d]_G}$ at η_C

Theorem 6. *Suppose*

(H1) $(\mathcal{P}_{G/H})^{[d]} \rightarrow U^{(d)}$ *is unramified at η_C ; and*

(H2) $\mathcal{P}^{[d]_H} \rightarrow (\mathcal{P}_{G/H})^{(d)}$ is unramified at $\eta_{C'}$.

Then $\mathcal{P}^{[d]_G} \rightarrow U^{(d)}$ is unramified at η_C .

Proof 1

Proof. We translate the unramifiedness conditions into the language of function fields and valuation rings. Let $K = K(C^{(d)})$ and $K' = K(C'^{(d)})$ be the function fields of the base symmetric powers. The generic points η_C and $\eta_{C'}$ of the respective exceptional divisors correspond to discrete valuations v_C on K and $v_{C'}$ on K' . The rational map f_X induces an inclusion $K \hookrightarrow K'$, with $v_{C'}$ extending v_C .

Let $M = K((\mathcal{P}_{G/H})^{[d]})$ and $L = K(\mathcal{P}^{[d]_G})$. From the torsor tower and the Cartesian square of [Proposition 4](#), we have field inclusions $K \subset M \subset L$ and $M \subset K'$, with $K(\mathcal{P}^{[d]_H})$ identified as the compositum $L \cdot K'$.

Let w_H be a valuation on $L \cdot K'$ extending $v_{C'}$. This naturally induces valuations lower in the tower: $w_L = w_H|_L$ and $w_M = w_H|_M$. Because the ramification index e is multiplicative in towers of finite extensions, we can compute $e(w_H|w_M)$ in two different ways: through K' and through L .

First, consider the tower $L \cdot K'/K'/M$. By hypothesis (H2), the extension $\mathcal{P}^{[d]_H} \rightarrow (\mathcal{P}_{G/H})^{(d)}$ is unramified at $\eta_{C'}$, which means $e(w_H|v_{C'}) = 1$. Furthermore, the map $q: (\mathcal{P}_{G/H})^{(d)} \rightarrow (\mathcal{P}_{G/H})^{[d]}$ is a quotient by the group $K_{G/H}$. Because the initial torsor $\mathcal{P} \rightarrow U$ has ramification bounded by d at p_∞ , the action of $K_{G/H}$ is free at $\eta_{C'}$. This ensures K'/M is unramified at w_M , meaning $e(v_{C'}|w_M) = 1$. Multiplying these gives:

$$e(w_H|w_M) = e(w_H|v_{C'}) \cdot e(v_{C'}|w_M) = 1 \cdot 1 = 1.$$

Second, consider the tower $L \cdot K'/L/M$. We have:

$$e(w_H|w_M) = e(w_H|w_L) \cdot e(w_L|w_M).$$

Since we established that $e(w_H|w_M) = 1$, and ramification indices are positive integers, it immediately forces $e(w_L|w_M) = 1$.

Finally, to show that $\mathcal{P}^{[d]_G} \rightarrow U^{(d)}$ is unramified at η_C , we must show $e(w_L|v_C) = 1$. By hypothesis (H1), the extension $(\mathcal{P}_{G/H})^{[d]} \rightarrow U^{(d)}$ is unramified at η_C , which gives $e(w_M|v_C) = 1$. Using multiplicativity one last time:

$$e(w_L|v_C) = e(w_L|w_M) \cdot e(w_M|v_C) = 1 \cdot 1 = 1.$$

This completes the proof. □

Proof 2

Proof. We translate the unramifiedness conditions into the language of function fields and valuation rings. Let $K = K(C^{(d)})$ and $K' = K(C'^{(d)})$ be the function fields of the base symmetric powers. The generic points η_C and $\eta_{C'}$ of the respective exceptional divisors correspond to discrete valuation rings $V_C \subset K$ and $V_{C'} \subset K'$. The rational map f_X induces an extension of these DVRs, $V_C \hookrightarrow V_{C'}$.

Let $L = K(\mathcal{P}^{[d]_G})$ and $M = K((\mathcal{P}_{G/H})^{[d]})$. The sequence of torsors $\mathcal{P}^{[d]_G} \xrightarrow{H} (\mathcal{P}_{G/H})^{[d]} \xrightarrow{G/H} U^{(d)}$ yields a tower of Galois extensions of function fields $L/M/K$, with $\text{Gal}(L/M) \cong H$ and $\text{Gal}(M/K) \cong G/H$.

By hypothesis (H1), the extension M/K is unramified at V_C . Let $W_M \subset M$ be a valuation ring extending V_C ; then $e(W_M/V_C) = 1$. To prove that $\mathcal{P}^{[d]G} \rightarrow U^{(d)}$ is unramified at η_C , it therefore suffices to show that the extension L/M is unramified at W_M .

Let $I_L \subseteq \text{Gal}(L/M) \cong H$ denote the inertia group of L/M at a valuation w extending W_M . We wish to show $I_L = \{1\}$.

From the Cartesian square of [Proposition 4](#), which holds over $U^{(d)}$ and thus induces a tensor product of function fields, we have

$$L_H := K(\mathcal{P}^{[d]H}) \cong L \otimes_M K'.$$

maybe its true (with enough context, but not right away) This means L_H is the compositum $L \cdot K'$. Because L/M is a Galois extension, L and K' are linearly disjoint over M . Consequently, $\text{Gal}(L_H/M) \cong \text{Gal}(L/M) \times \text{Gal}(K'/M)$.

Let w' be a valuation of L_H extending w on L and $V_{C'}$ on K' , and let $I_{L_H} \subseteq \text{Gal}(L_H/M)$ be its inertia group. The projection maps

$$p_1 : \text{Gal}(L_H/M) \rightarrow \text{Gal}(L/M) \cong H, \quad p_2 : \text{Gal}(L_H/M) \rightarrow \text{Gal}(K'/M)$$

restrict to the inertia groups, giving $p_1(I_{L_H}) = I_L$.

The inertia group of the extension L_H/K' at w' is exactly $\ker(p_2) \cap I_{L_H}$. By hypothesis (H2), the extension $\mathcal{P}^{[d]H} \rightarrow (\mathcal{P}_{G/H})^{(d)}$ is unramified at $\eta_{C'}$, meaning L_H/K' is unramified at $V_{C'}$. Therefore,

$$\ker(p_2) \cap I_{L_H} = \{1\}.$$

This trivial intersection implies that the restriction $p_2|_{I_{L_H}} : I_{L_H} \rightarrow \text{Gal}(K'/M)$ is injective.

At the same time, the map $q : (\mathcal{P}_{G/H})^{(d)} \rightarrow (\mathcal{P}_{G/H})^{[d]}$ identifies $\text{Gal}(K'/M)$ with the quotient group $K_G/K_H \cong K_{G/H}$. Crucially, the initial assumption that the torsor $\mathcal{P} \rightarrow U$ has ramification bounded by d at p_∞ ensures that the action of $K_{G/H}$ on $C'^{(d)}$ is free at the generic point $\eta_{C'}$ of the exceptional divisor. Thus, the extension K'/M is itself unramified at W_M , meaning its inertia group $p_2(I_{L_H}) \subseteq K_{G/H}$ is trivial.

Since $p_2|_{I_{L_H}}$ is injective and its image is trivial, we conclude $I_{L_H} = \{1\}$. Finally, since $I_L = p_1(I_{L_H})$, it follows that $I_L = \{1\}$, so L/M is unramified at W_M . This completes the proof. $\square$

References

- [Gui19] Quentin Guignard. “On the ramified class field theory of relative curves”. In: *Algebra & Number Theory* 13 (May 2019). Revised version submitted on 29 May 2019, pp. 1299–1326. DOI: [10.2140/ant.2019.13.1299](https://doi.org/10.2140/ant.2019.13.1299). arXiv: [1804.02243](https://arxiv.org/abs/1804.02243) [math.AG].